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COMET FWC 071

Wi-Fi Connectivity Testing using VAMAN ESP32 Board

1 Objective

The objective of this experiment is to verify the Wi-Fi functionality of the VAMAN ESP32 development board by connecting it to a wireless access point, obtaining an IP address from the router, and monitoring the communication through the serial terminal.

2 Hardware Requirements

1. VAMAN ESP32 Development Board
2. USB Type-C Cable
3. Ubuntu 24.04 System
4. PlatformIO IDE
5. Wi-Fi Router / Mobile Hotspot

3 Software Requirements

1. Ubuntu 24.04 LTS
2. PlatformIO
3. Arduino Framework
4. ESP32 Board Package
5. Serial Monitor (115200 baud)

4 Program Description

The program initializes the serial communication at 115200 baud and configures the ESP32 in Station (STA) mode. The board attempts to connect to the specified Wi-Fi network using the provided SSID and password.

A maximum of twenty connection attempts are made. If the connection is successful, the assigned IPv4 address is displayed on the serial monitor. Otherwise, an error message along with the Wi-Fi status code is printed.

After successful connection, the ESP32 continuously prints a heartbeat message indicating that the controller is running properly.

5 Algorithm

1. Initialize Serial Communication.
2. Configure ESP32 in Station Mode.
3. Start Wi-Fi Connection.
4. Wait until Wi-Fi is Connected.
5. If Connected
   (a) Print "WiFi Connected".
   (b) Display Assigned IP Address.
6. Else
   (a) Print Connection Failed.
   (b) Display Wi-Fi Status Code.
7. Continuously print system running status every two seconds.

6 Source Code

```
#include <Arduino.h>
#include <WiFi.h>

const char* ssid = "vamsi";
const char* password = "123456789";

void setup()
{
    Serial.begin(115200);

    delay(3000);

    Serial.println("ESP32 WiFi Test Started");

    WiFi.mode(WIFI_STA);

    WiFi.begin(ssid,password);

    int retry = 0;

    while(WiFi.status()!=WL_CONNECTED && retry<20)
    {
        Serial.print(".");
        delay(500);
        retry++;
    }

    if(WiFi.status()==WL_CONNECTED)
    {
        Serial.println("WiFi Connected!");
        Serial.print("IP Address : ");
        Serial.println(WiFi.localIP());
    }
    else
    {
        Serial.println("WiFi Connection Failed!");
        Serial.println(WiFi.status());
    }
}

void loop()
{
    Serial.println("ESP32 is running...");
    delay(2000);
}
```

7 Testing Procedure

1. Connect the VAMAN ESP32 board to the Ubuntu system using USB.
2. Open the project in PlatformIO.
3. Build and upload the program.
4. Open the Serial Monitor at 115200 baud.
5. Power on the Wi-Fi router/mobile hotspot.
6. Verify that the SSID and password match the program.
7. Observe the connection status on the serial monitor.
8. Verify that the ESP32 receives a valid IP address.
9. Confirm periodic status messages after successful connection.

8 Expected Serial Output

```
==================================
ESP32 WiFi Test Started
==================================

Connecting to WiFi...
......

WiFi Connected!
IP Address: 192.168.29.105

ESP32 is running...
ESP32 is running...
ESP32 is running...
```

9 Observation

- The ESP32 successfully entered Station Mode.
- The ESP32 was taking the code and entrring into download mode after resetting the esp32 is unable to generate the IP ADDRESS.
- The board is not connected to the configured Wi-Fi network.

10 Result

The Wi-Fi interface of the VAMAN ESP32 board was successfully tested. The ESP32 established a reliable connection with the configured wireless network, obtained an IP address from the DHCP server, and continuously executed the application without communication failure. The experiment verified the proper operation of the ESP32 wireless subsystem and its readiness for future IoT and networking applications.